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Pervasive Remagnetization of Paleozoic Rocks Acquired at the Time of Mesozoic Folding in the South China Block

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A total of 152 paleomagnetic samples was collected from 23 sites distributed over six Paleozoic formations, with ages ranging between Middle Ordovician and Early Permian, in the Tangshan type section in the Nanjing area of South China. Stepwise thermal demagnetization has successfully resolved two magnetization components from these samples. Component A with northerly shallow tilt-corrected directions resides in approximately 95% of the total number of samples analyzed; this component passes fold tests at the 99% significance level in three formations. The overall mean direction of component A is 350/-3 (tilt-corrected, a95=7.2). Component B, isolated from 19 samples in two formations (C1 and D3), yields negative fold tests at the 95% significance level and is, therefore, younger than the Mesozoic folding. The overall site mean direction of component B is 354/-4 (in situ, a95=8.4). The tilt-corrected direction of component A clearly resembles the in situ direction of component B, indicating that both components are likely to be of approximately the same age. We infer that they are both remagnetizations, associated with the deformation. A review of the Paleozoic paleomagnetic data reported by other authors from adjacent areas shows that similar remagnetization situations may prevail. It is concluded that the northeast part of south China has experienced pervasive remagnetizations at the time of folding. The direction of this folding-induced remagnetization yields a new well-defined paleopole position (56°N, 314°E) for south China. Although this paleopole is significantly different from Mesozoic poles reported to date, we infer that a Triassic to Jurassic age is most likely for this remagnetization.

INTRODUCTION

After the introduction of the concepts of primary and secondary magnetization [e.g., *Irving*, 1964; *Creer*, 1968], remagnetization of sedimentary rocks has long been an important subject of study in paleomagnetism. Research conducted in the 1980s has yielded unequivocal evidence that sedimentary rocks over a large portion of interior cratonic North A merica have been remagnetized during the Pennsylvanian and Permian period [*McCabe and Elmore*, 1989]. A comparison of reviews of Paleozoic data for North A merica by *Van der Voo* [1981, 1989] indicates that several of the paleopoles from pre-Pennsylvanian rocks cited by *Van der Voo* [1981] as "reliable" had been shown to be based on late Paleozoic remagnetizations by the time of his 1989 review. Evidence for such a remagnetization has included (1) the resemblance of the paleopole positions from pre-Pennsylvanian rocks to those with Permocarboniferous ages, (2) the negative or syn-folding fold tests and (3) observations of apparently secondary features of magnetic minerals residing in a variety of sedimentary rocks [*McCabe et al*, 1989, 1984, 1983; *Miller and Kent*, 1988; *Kent*, 1985; *Bachtadse et al.*, 1987; *Suk et al.*, 1990*a*, 1990*b*; *Jackson.*, 1990].

Such remagnetization phenomena have been documented not only from North America but also from a large portion of the rest of the world, such as Europe [*Gehring and Heller*, 1989], Australia [*Li et al.*, 1989], as well as East Asia [Chen and Courtillot, 1990]. The recognition of remagnetizations in ancient rocks has in fact become an important revolution in paleomagnetism.

The South China Block is one of the major tectonic units of Asia. Despite numerous investigations spanning more than a decade, the paleomagnetic data base for the Paleozoic of South China remains sparse and the subject controversial. A major cause for the slow progress has been the suspicion, although not often openly stated, that remagnetization in South China may be as prevalent as in North America or in the other parts of the world mentioned above. Several Paleozoic sedimentary formations exposed in the vicinities of Yichang in Hubei Province and Nanjing in Jiangsu Province were studied by *Kent et al.* [1987], who concluded that widespread late Mesozoic to recent remagnetization occurred there. Thus a possible remagnetization problem in South China has already been noted by paleomagnetists.

Remagnetization in general can be classified as post-folding when it can be characterized by a negative fold test or as syn-folding when a certain percentage of unfolding exists at which the precision parameter reaches a maximum and is significantly different from that at either zero or one hundred percent. A remagnetization is pre-folding when it can be demonstrated, [e.g., *Hodych*, 1989], that a positive fold test is accompanied by a negative conglomerate or contact test. Remagnetization events have been commonly considered to be associated with orogenies and deformation [*McCabe et al.*, 1989, 1984, 1983; *Jackson*, 1990; *Kent*, 1985]. It appears that an ancient rock may acquire a remagnetization at any "moment" during an orogeny as long as favorable chemical and physical conditions occur. At the two extreme ends of the spectrum of possibilities, it is thus possible that a remagnetization component was introduced at an early or late stage, respectively, of a folding process [e.g., *Miller and Kent*, 1986, 1988]. Logically, a component acquired during the early stages of folding will appear to pass the fold test, whereas one acquired at later stages of the folding will fail the test. Regardless of this apparently black-or-white situation, both magnetizations are, in fact, remagnetizations associated in time with the deformation.

In an effort to augment the sparse Paleozoic data base for South China, 152 oriented samples were analyzed for paleomagnetic study from 6 Paleozoic sedimentary formations exposed in the Tangshan section near Nanjing City in Jiangsu Province. Evidence for a pervasive folding-induced remagnetization has been found in these samples. A review of other paleomagnetic data from several Paleozoic rocks in the vicinity of Nanjing investigated by *Zhao* [1987] and from Devonian red beds in south Jiangsu Province, [*Zhai*, 1984; *Zhai*

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et al., 1987], shows that in all these localities similar remagnetization situations prevail.

GEOLOGICAL SETTING AND SAMPLING

An outline of the geological development of China in terms of plate tectonics was given by *Zhang et al.* [1984]. The northern part of the South China Block was fundamentally consolidated after the Yangtze Orogeny (ca. 850 Ma). The Precambrian basement is overlain by a thick sedimentary cover. The Nanjing area, located in the northeast part of South China, about 100 km south to the Dabi Mountains which are the eastern extension of the Qinlin Orogenic Belt (Figure 1), exposes geologically well-studied Tangshan Paleozoic and Triassic type sections. It remained a stable platform experiencing only minor vertical movements that are reflected, for instance, in depositional hiatuses such as during the Early and Middle Devonian. In this area, folding produced tight upright to overturned anticlines and more open synclines which contain Triassic strata in their cores. The major folding phase therefore likely occurred in Triassic or Jurassic time [*Jiangsu and Shanghai Regional Stratigraphic Tables Compilation*, 1978], associated with the collision of the South China and North China blocks along the Qinling suture roughly at the same time. Subsequent intrusive activity happened about 116 Ma [*Chinese Academy of Geological Sciences*, 1973], corresponding to the Yangshanian Orogeny.

A total of 152 oriented drill-core samples was analyzed in this study, distributed over 23 sites in six formations in Tangshan sections with ages ranging from Middle Ordovician through Early Permian. Sites are groupings of five or more independently oriented samples taken stratigraphically over several meters. Bedding dips range from about 60 degrees to nearly vertical to overturned and generally are steep. Different limbs of folds could be sampled in the Upper Devonian and Lower Carboniferous formations and both overturned and right-side up Silurian localities were included. Although only one site could be sampled at one of the limbs for the Devonian and Carboniferous formations due to limited exposed area and drilling difficulties in the field, fold tests could still be performed on each of these three formations. All 2.5cm diameter samples drilled in the field were cut into specimens of 2.2cm height. All 23 sites of the sampled formations are listed in Table 1 with dominant lithologies, ages and number of samples, structural attitudes and their paleomagnetic results which are discussed below.

PALEOMAGNETIC MEASUREMENTS AND RESULTS

Measurements of the Natural Remanent Magnetization (NRM) were carried out on a ScT cryogenic magnetometer, with AF and thermal demagnetizations performed on Schonstedt equipment at the University of Michigan, in a shielded room with rest field generally less than 200nT. Demagnetization results have been plotted in orthogonal vector diagrams [*Zijderveld*, 1967] and in stereographic projections from which components of magnetization were recognized, followed by Principal Component Analysis [*Kirschvink*, 1980]. The NRM intensities of the samples ranged between 0.1 and 5 mA/m, with no significant correlation between intensities and the rock types. AF treatment did not succeed in reducing the intensity to a small percentage of the initial NRM, so stepwise thermal demagnetization was used on the majority of samples. The results described below for the Tangshan sections are summarized in Table 1.

A rather straightforward demagnetization behavior has been found in all Lower Permian and Upper Carboniferous samples. After removal of a spurious magnetization in the initial stages of demagnetization, the dominant magnetization component, labeled component A, is revealed as northeasterly and steep in situ (Figures 2*a* and 2*b*) and northerly with shallow inclination

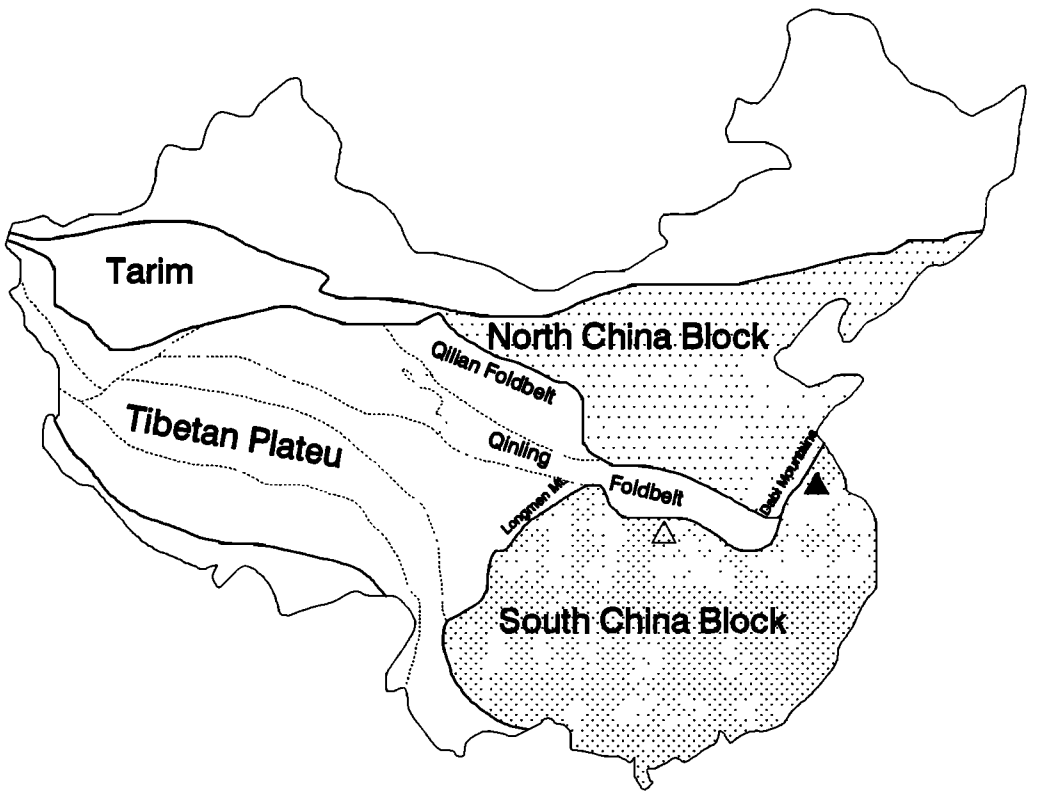


Fig. 1. Tectonic sketch map of China after *Zhang et al.* [1984]. Solid triangle is the locality of the Tangshan section. Open Triangle is the locality of Yichang where a paleomagnetic study has been carried out by *Kent et al.* [1987].

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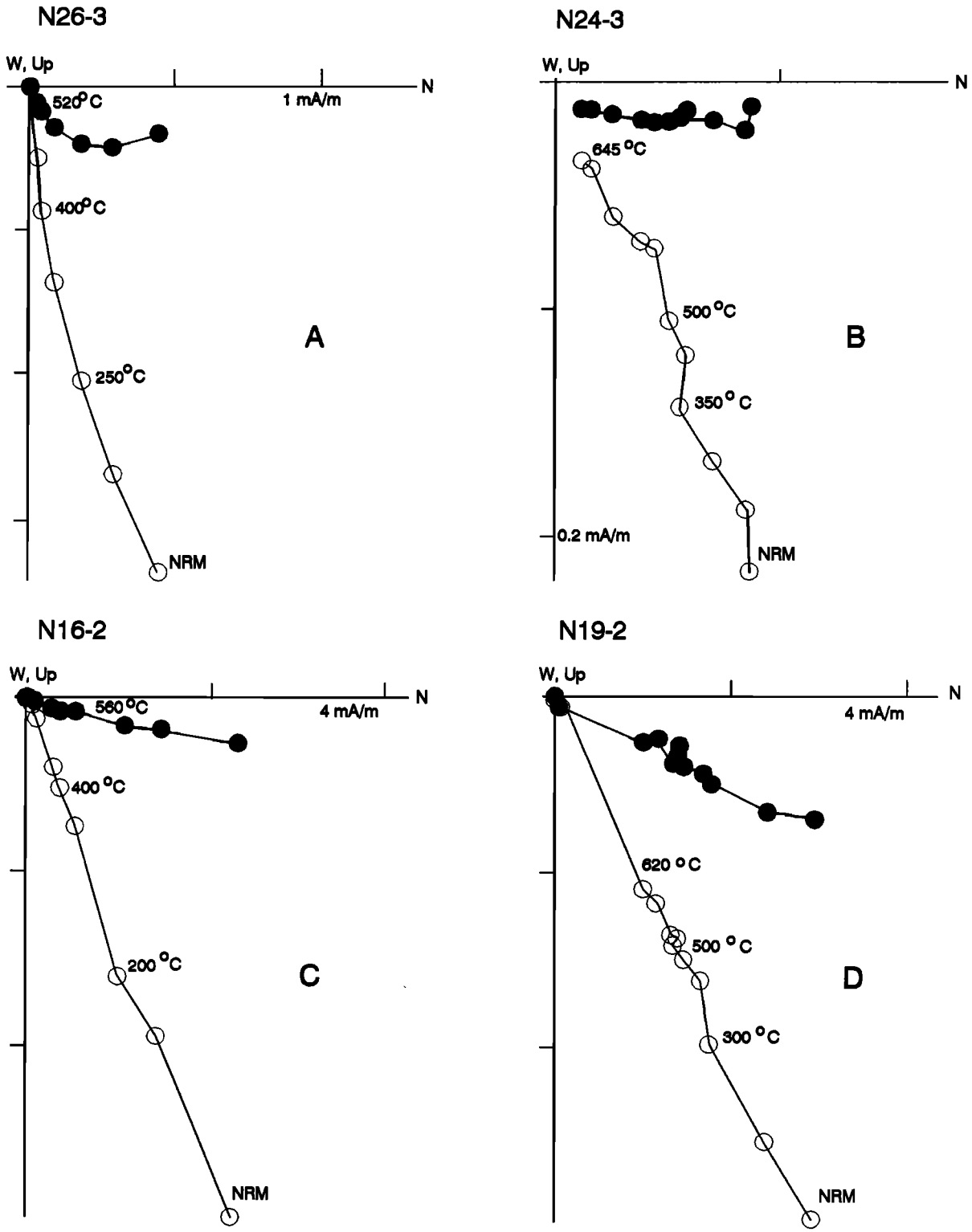


Fig.2. Orthogonal plots of demagnetization data representative of formations of (*a*) Early Permian, (*b*) Late Carboniferous, and (*c*) and (*d*) Early Carboniferous ages in in-situ coordinates. Open (solid) circles represent end points plotted in vertical (horizontal) projection. Numbers with degrees adjacent to data points are thermal demagnetization levels in degree Celcius.

after correction for the bedding tilt. The in-situ direction is much steeper than the present-day field and, therefore, appears to represent an ancient magnetization. The wide range of unblocking temperatures (generally 400 °C to 675 °C) implies that this component resides both in magnetite and hematite in these units.

There is more variability in the magnetization of the Lower Carboniferous and Upper Devonian formations. Some of the samples (e.g., all samples in site 16) have an essentially univectorial NRM with consistent directions corresponding to component A as defined above (Figures. 2*c and* 2*d*). However, another component with northerly or northwesterly shallow in-

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situ direction has been successfully isolated from the rest of the samples (Figures 3 and 4). A distinctive feature is the presence of both polarities (1) for the B component as revealed in demagnetization of the red Upper Devonian sandstone and the gray Lower Carboniferous dolomite, and (2) for component A in the Upper Devonian formation. Two polarities are rarely seen in the same sample but can occur in different samples from the same site; the relative thermal stabilities of both component A and B are otherwise independent of polarity, showing broad unblocking temperature ranges up to 670°C. It is important to note that the B component in these samples co-exists with component A (Figures 3 and 4), indicating that at least one of

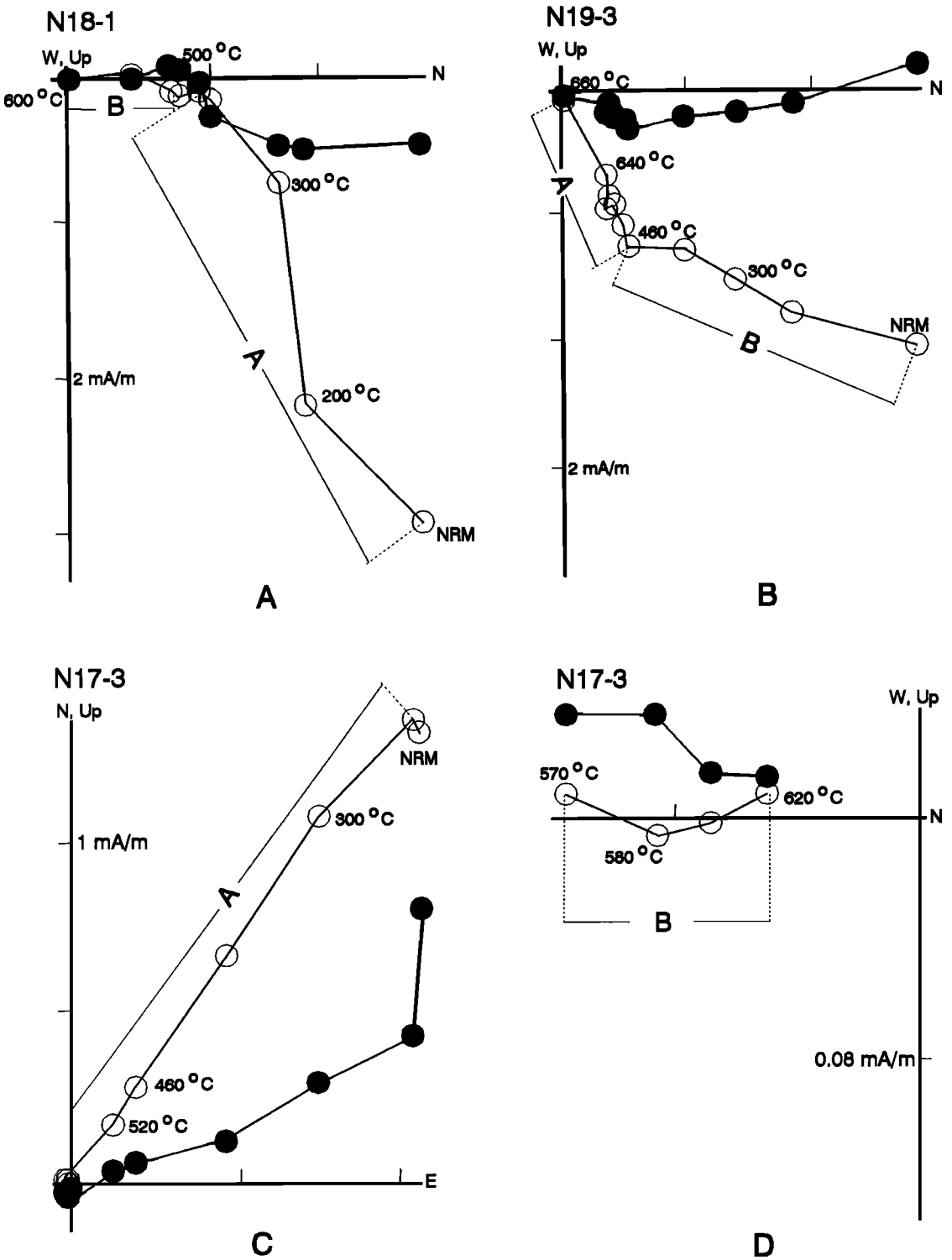


Fig.3. Orthogonal plots of demagnetization data representative of part of the Lower Carboniferous Laohushan formation in which both components, A and B, have been observed. Plotted in geographic coordinates, with Figure 3*d* showing a reoriented enlargement of the higher-temperature portion of the diagram in Figure 3*c*. Component A is more stable in N19-3, but less stable in N18-1 and N17-3, than component B. Conventions as in Figure 2.

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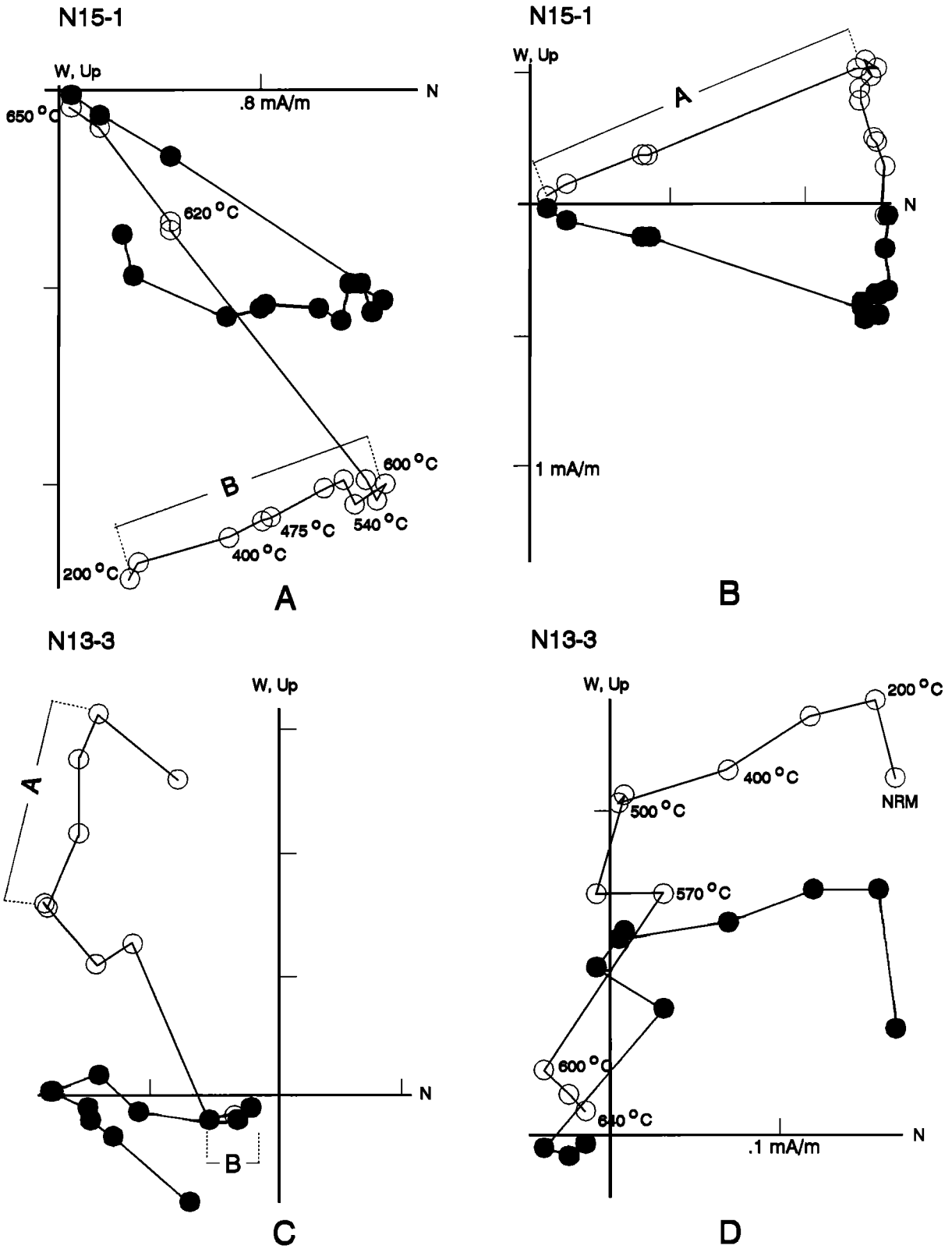


Fig.4. Orthogonal plots of demagnetization data representative of the Devonian Wutong formation in (*a*) geographic and (*b*) stratigraphic coordinates, showing both components A and B residing in an individual sample (N15-1). The in-situ direction of component B with lower blocking temperatures (Figure 4*a*) is antipodal to the tilt-corrected direction of component A (Figure 4*b*). Component B displays reversed polarity (Figures 4*a* and 4*c*). Figures 4*e* and 4*f* show the demagnetization pattern of a sample carrying component A with a reversed polarity in (*e*) geographic and (*f*) bedding coordinates, respectively. Other conventions as in Figure 2.

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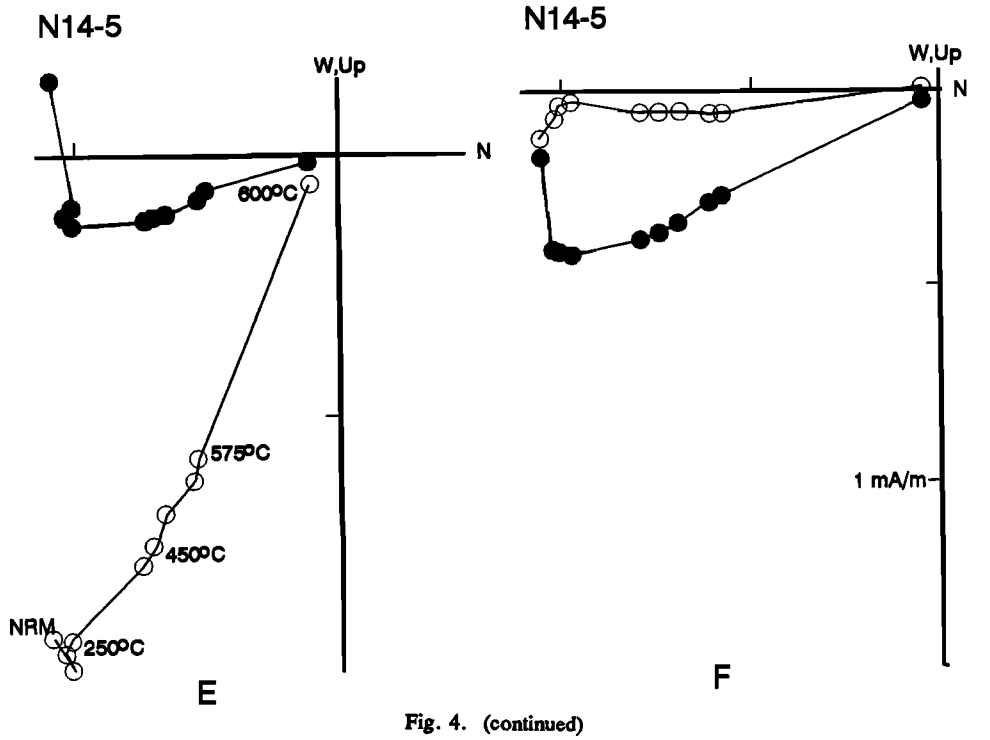


Fig. 4. (continued)

them is secondary. The relative unblocking temperatures of the A and B components may nevertheless vary for different samples. In the Devonian samples, the unblocking temperatures of component B generally exceed 590° (Figure 4), indicating that it may well be carried by hematite. On the other hand, the stability of component A is more complex with unblocking temperatures ranging between 200°C and 670°C, showing that it might reside both in magnetite and hematite. It is clear that the unblocking temperature of component A is higher in sample N19-3 (Figure 3*b*) but lower in samples N18-1 and N17-3 (Figures 3*a*, 3*c* and 3*d*) than that of component B. A similar situation can also be seen in Figure 4, where the unblocking temperature of the A

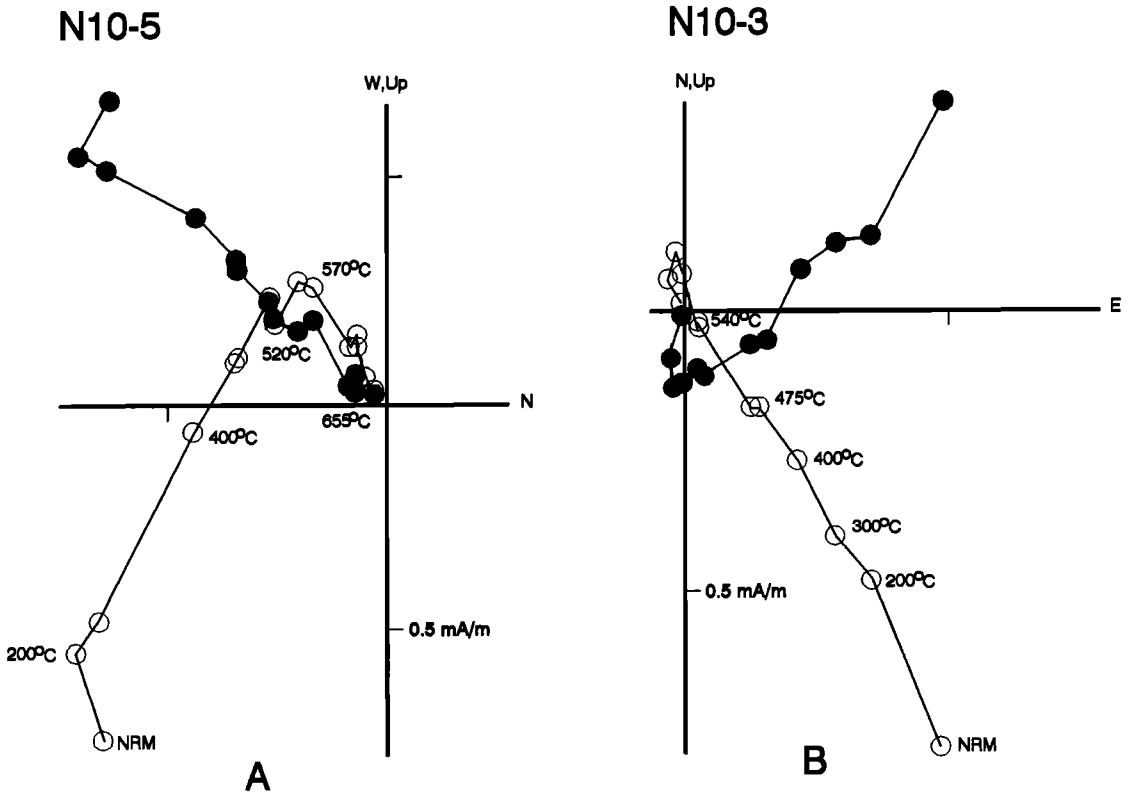


Fig.5. Orthogonal plots of demagnetization data representative of Silurian sandstones. Component A, displaying reversed direction, is shown in both samples at higher temperatures. Other components seen from the trajectories have random directions. Conventions as in Figures 2 and 3.

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component is higher in sample N15-1 but lower in 13-3. In both formations, therefore, we can have unblocking temperatures of component A that are higher as well as lower than that of component B; moreover, components A and B can both be of either normal or reversed polarity (e.g., Figures 4*a*, 4*b* for component B and Figure 4*f* for component A).

In a mauve-colored Middle/Upper Silurian sandstone unit, only component A seems to be present, after aberrant lower-blocking temperature directions have been removed (Figure 5*a*). The predominant polarity of component A in this unit appears to be reversed, but a normal direction superposed on a reversed one is perhaps seen in a few samples (Figure 5*b*). In terms of unblocking temperature, both magnetite and hematite again are major contributors to the remanence in this unit.

The demagnetization behavior of the samples from the Middle Ordovician unit is less remarkable. A characteristic component with the tilt-corrected direction of component A is revealed in two of the four sites after a soft present day field (pdf) overprint is removed (Figure 6*a*). The remaining two sites seem to display only a pdf direction (Figure 6*b*), although this is somewhat ambigous when considering the similar northerly and shallow directions of all four sites after tilt correction. The unblocking temperatures of magnetization in this unit are no higher than 570°C, indicating that the remanence is carried only by magnetite.

ANALYSIS OF RESULTS

About 90% of the samples provided interpretable demagnetization trajectories. In the remaining samples, progressive demagnetization generally did not reveal any systematic behavior, usually because of instability of magnetization but frequently also because of weak NRM intensities. The site and formation means and associated statistical parameters are listed in Table 1.

The available data calculated from both limbs of a fold allow fold tests for three formations, i.e., Lower Carboniferous, Upper Devonian and Middle/Upper Silurian ages. With *McElhinny's* [1964] fold test, positive tests were obtained for component A from all three formations at the 99% confidence level; the *McFadden* [1990] fold test statistic ξ demonstrate that the hypothesis that these magnetizations were acquired in-situ can be rejected at the 95% confidence level. In contrast, the fold tests on component B in the Lower Carboniferous and Upper Devonian units are significantly negative at the 95% confidence level provided that certain polarity conditions are met as will be detailed below. It is thus apparent that component A was acquired during some period prior to the folding, whereas component B is younger than the folding in these cases. There are no fold tests available for the remaining formations, but the resemblance of their characteristic directions to the directions that pass the fold tests evidently indicates that component A is documented in all the units studied here and is prefolding in age.

The sample directions of component A are plotted in the stereo net of Figure 7. The tilt-corrected declinations of component A display some scattering, represented by northwesterly-southeasterly as well northerly directions. The vertical to overturned attitudes of the Silurian formations may be the possible cause for this inconsistency, as correction for very steep dips in the case of folds with even a minor plunge may cause "apparent" rotations [*McDonald*, 1980].

The fold test for component B (and to some extent also for component A) hinges on a critical consideration which involves the polarity choice for the directions of the magnetizations. As noted earlier, the A and B components show both polarities, and this applies to in-situ and tilt-corrected coordinates (Figures 7 and 8). The statistical parameters (e.g., k1 and k2 as the precision parameters before and after un folding, respectively) critically depend on whether a given direction is used as observed or whether it is inverted to the opposite polarity in the

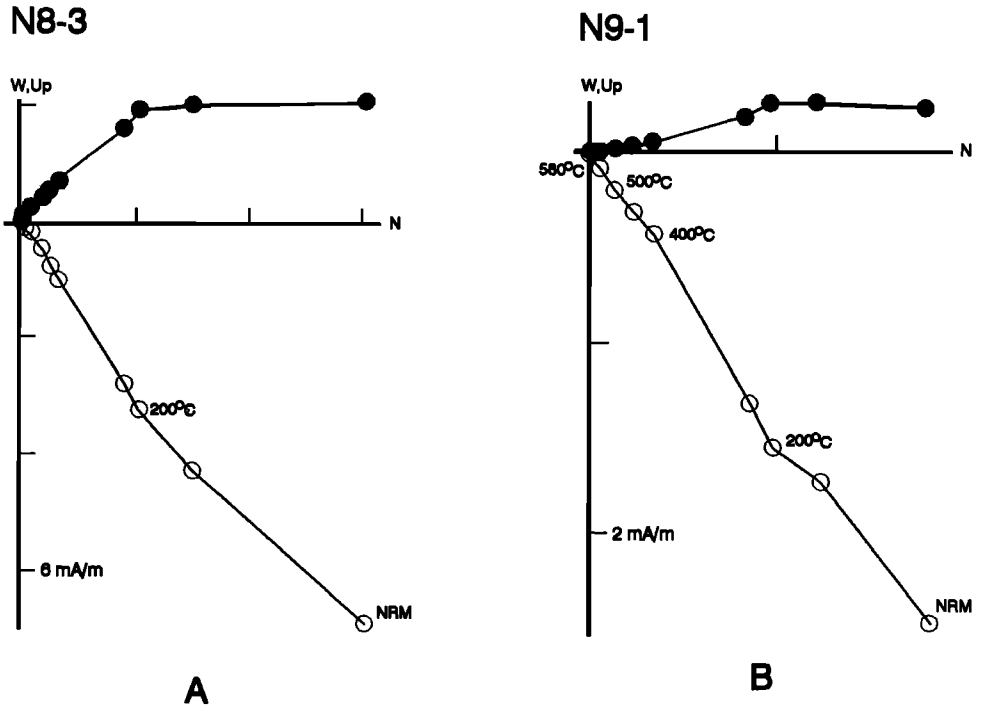


Fig.6. Orthogonal plots of demagnetization data representative of the Middle Ordovician Baota formation. Component A is revealed in two of the four sites (Figure 6*a*). It is not clear whether the pdf overprint is successfully removed in the samples from the other two sites (Figure 6*b*). Conventions as in Figure 2.

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TABLE 1. Paleomagnetic Results From Tangshan Section (Location: 32°N, 119°E)

Site	Strike/Dip	n/N	In Situ		Tilt-Corrected		k2	α_{95}	k2/k1	ξI	ξC
			D	I	D	I					
Lower Permian Qisha Limestone											
26	255/77	6/8	48	79	357	7	218.3	4.5			
27	255/77	8/8	39	81	352	8	39.7	9.1			
28	255/77	5/5	26	71	358	-2	132.8	6.7			
Mean Qisha		3/3	35	77	355	-5	188.8	9.0			
Upper Carboniferous Chuanshan Limestone											
22	255/77	4/6	5	57	353	-9	97.4	9.4			
23	255/77	7/7	31	65	3	-4	206.6	4.2			
24	255/77	8/8	27	65	1	-6	85.1	6.0			
25	255/77	6/6	19	72	355	-2	377.5	3.5			
Mean Chuanshan		4/4	21	68	358	-5	210.8	6.3			
Lower Carboniferous Laohushan Dolomite, Component A											
16	256/84	9/9	14	67	357	-14	313.0	2.9			
17	65/65	6/9	50	-52	12	-11	75.0	7.8			
18	255/77	5/6	32	67	1	-3	94.6	7.9			
19	255/77	7/9	22	63	1	-9	327.3	3.3			
20	255/77	6/8	28	72	357	-1	108.3	6.5			
21	255/77	7/8	34	72	358	-1	122.7	5.5			
Mean Laohushan Comp. A		6/6	14	77	1	-7	104.8	6.6	32.05	3.05	1.35
Lower Carboniferous Laohushan Dolomite, Component B											
17	65/65	5/9	348	-1	4	63	9.2	26.0			
18-20	255/77	6/23	346	-7	35	-83	12.1	20.2			
Mean Laohushan Comp. B		11/32	347	-3	17	-87	6.9	19.0	0.51	0.74	10.34
Upper Devonian Wutong Sandstone, Component A											
13	56/60	5/6	25	-69	347	-17	12.0	23.1			
14	59/60	6/6	170	67	159	9	386.0	3.4			
15	256/84	7/7	35	60	8	13	27.6	11.7			
Mean Wutong Comp. A		3/3	326	-82	351	-13	29.9	24.7	3.83	2.29	0.14
Upper Devonian Wutong Sandstone, Component B											
Mean Wutong Comp. B		8/19	188	-5	167	-67	4.7	28.6	0.39	2.01	3.71
Middle/Upper Silurian Fentou Sandstone											
10	72/128	7/7	200	-44	316	-1	36.5	10.1			
11	72/128	3/5	12	52	144	-4	42.2	19.2			
12	58/87	4/4	83	77	137	-3	47.1	13.5			
Mean Fentou		14/16	25	60	139	-2	80.6	4.5	5.40	5.40	0.58
Middle Ordovician Baota Limestone											
O1	290/77	5/5	307	42	355	-4	113.0	7.2			
O2	290/77	4/5	313	49	342	-5	141.0	7.2			
O3*	290/77	5/5	353	54	4	-19	741.0	2.8			
O4*	290/77	5/5	351	56	358	-16	208.3	5.3			
Mean Baota		9/10	309	45	338	-4	141.6	4.3			

ξI and ξC are the *Mcfadden* [1990] fold test statistic parameters, before and after correction for bedding tilt, respectively. D, declination; I, inclination.
* pdf directions.

process, and whether this is done in in-situ or tilt-corrected coordinates. Because of the steep, northerly or southerly facing dips (Table 1), directions of one polarity are corrected into two approximately antipodal groups. The in-situ directions of the opposite polarity are also corrected into two approximately antipodal directions. Hence, the two polarities exist in both in-situ and tilt corrected coordinates. However, as we will show in a later section, there are results from an adjacent area [from *Zhai*, 1984] which do not suffer from this ambiguity. We are confident, therefore, that our interpretation of the A and B components has validity.

The overall site mean directions of both components A and B, regardless of the rock ages, are summarized in Table 2. The correlation between the directions is rather striking and of special interest. The pre-folding component (A) and post-folding component (B) virtually display an identical direction, i.e., northerly and shallow, indicating that both components are records of similar geomagnetic directions. Considering that the Paleozoic formations in the Nanjing area were not folded until the Mesozoic, a plausible interpretation for this situation is that both components have been acquired more or less during the same period, i.e., the Triassic or Jurassic. In this case, component A must have been induced at a very early stage of the deformation while component B was acquired at a late stage, as illustrated in Fig. 9. This unique combination of components A and B is here called a "folding induced remagnetization". This terminology implies a difference from the concept of a "syn-folding" remagnetization, which refers only to a single component.

The Triassic-Jurassic paleopole positions (55/316 and 55/309 from prefolding and postfolding data respectively) for South China are herein determined in accordance with the directions of

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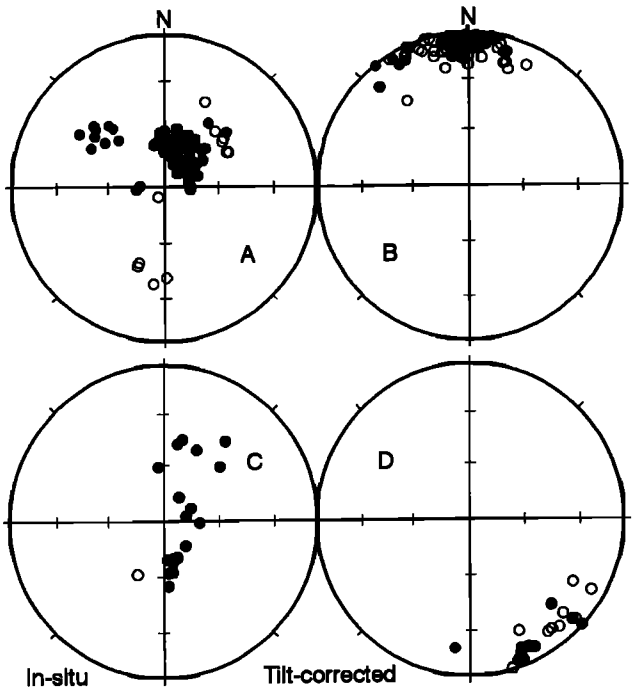


Fig.7. Equal-area projections of the sample directions of component A, as revealed from all formations, before (left) and after (after) correction for the tilt of the strata. Solid (open) circles represent directions plotted onto the lower (upper) hemisphere. Figures 7*a* and 7*b* show the directions with tilt-corrected normal polarity. Figures 7*c* and 7*d* show the directions with tilt-corrected reversed polarity.

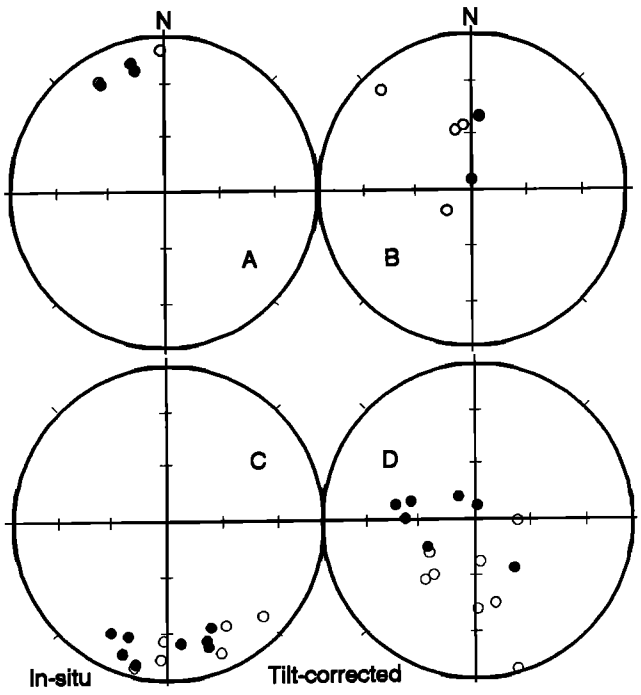


Fig.8. Equal-area projections, as in Figure7, of directions of component B before (Figures 8*a* and 8*c*) and after (8*b* and 8*d*) correction for the tilt of the strata. Figs. 8*a* and 8*b* show the directions with in-situ normal polarity. Figures. 8*c* and 8*d* are the directions with in-situ reversed polarity.

the folding-induced remagnetizations found in the Tangshan section. These two pole positions do not closely resemble any Mesozoic to Cenozoic paleopoles reported to date for South China [*Kent et al.*, 1987; *Lin*, 1984; *Otofuji et al.*, 1990]. An explanation might be linked with the possibility that our sampling area is rotated with respect to some or all of the other South China sampling areas. Such a rotation remains speculative because one needs precisely dated and contemporaneous magnetizations to determine its magnitude. More compelling arguments for the age of the early Mesozoic remagnetization will thus critically depend on the acquisition of more data. If the assignment of a Triassic-Jurassic age for our poles is proven to be valid, the tectonic evolutionary history of South China in the Mesozoic will have to be reevaluated.

Previous Results Showing Similar Behavior and Discussion

A Devonian red bed in south Jiangsu province, at a distance of 150 km from the Tangshan section, was studied by *Zhai* [1984]. The results have been reported as of pre-folding age with a mean direction similar to that of our folding induced remagnetizations [*Zhai et al.*, 1987]. Upon reviewing the relevant chapter in Zhai's thesis, we have found that actually two characteristic magnetization components are listed in the sample file, but that the second component (B) is not discussed. The statistical results of both A and B components are summarized in Table 3, where Jikang and Maoshan are localities on the two limbs of an anticline. Apparently, component A passes the fold test but component B fails (Figure 10). Comparison of these components A and B with those from our study indicates that a similar remagnetization situation apparently prevails in both the Tangshan section and South Jiangsu province. We note that the dips of the strata (Table 3) are not as steep as those in our own study and that, therefore, the polarity ambiguity (referred to earlier) does not exist here.

Paleomagnetic investigations of several Paleozoic formations exposed in the vicinity of Nanjing City were also conducted by *Kent et al.* [1987] and *Zhao* [1987]. Directions resembling our component A were obtained by *Kent et al.* [1987] from units of

TABLE 2. Paleomagnetic Results From Tangshan Section (Location: 32°N, 119°E)

Component	N	In-Situ		Tilt-Corrected		k	α_{95}	Paleopole		Dp	Dm
		D	I	D	I			Latiitude	Longitude		
A#	21	[10	74]	350	-3	18.0	7.2	55	316	3.6	7.2
B*	19	354	-4	[350	75]	14.6	8.4	55	309	4.3	8.4

Directions thought not to represent ancient directions are given in brackets.
Caculated from site means; α_{95} is given after tilt correction.
* Calculated from sample directions; α_{95} is given in situ.

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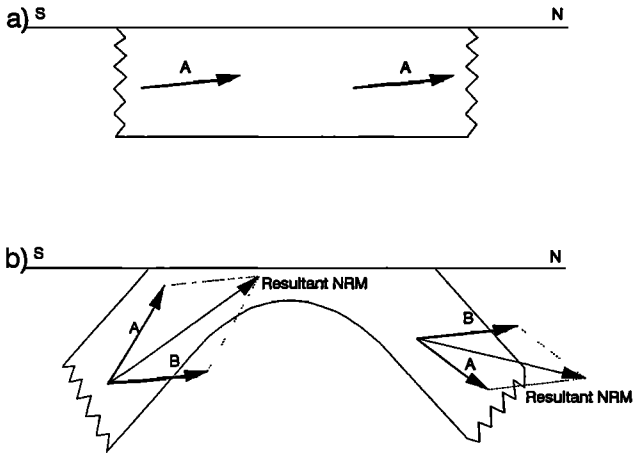


Fig.9. Illustration of the directional correlation between two components acquired at the early and late stage of folding respectively. (*a*) component A acquired at the early stage of folding and (*b*) component B acquired at the late stage of folding. The directions of both components are different in the same coordinates but the in-situ direction of component B resembles the tilt-corrected direction of component A.

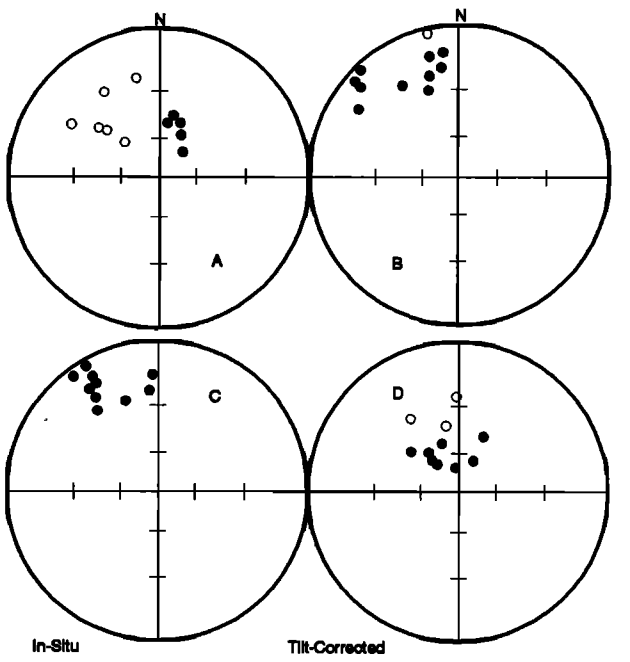


Fig.10. Equal-angle projections of the sample directions of component A passing the fold test (Figures 10*a* and 10*b*) and component B failing the fold test (Figures 10*c* and 10*d*), as revealed from Devonian red beds in South Jiangsu Province, before (Figures 10*a* and 10*c*) and after (Figures 10*b* and 10*d*) correction for the tilt of the strata. Conventions as in Figure 7. Data are reported by *Zhai* [1984].

Early and Late Carboniferous, Early Permian as well as Early Triassic ages. The directions may pre-date folding in the Lower Carboniferous and Triassic rocks while they postdate folding for the Permian rocks based on tilt tests. The test confidence levels, however, are apparently low. Their Devonian direction, moreover, is completely different from our observations (northeasterly steep upward in tilt-corrected coordinates). The cause for this inconsistence is not clear.

Zhao [1987] also reported several magnetization components in the Paleozoic (Silurian, Carboniferous and Permian) rocks from the areas near Nanjing. He grouped the components according to their blocking temperature (Tb) ranges. We recall, however, that for our study both hematite and magnetite were determined to be the carriers of the A and B components and that such a grouping by Tb would not have worked. In an attempt to test whether Zhao's results show similarities to ours, we have regrouped his results according to their directions (before or after tilt correction) and the outcome is presented in Table 4. Both A and B component directions are indeed observed (Figures 11 and 12) and component B convincingly fails the fold test. It can be argued, therefore, that all previous and present studies of the northeastern corner of South China Block reveal similar northerly shallow (or their antipodal) directions regardless of the Paleozoic age of rocks. While the A component appears to be

TABLE 3. Paleomagnetic Results From Maoshan and Jikang Red Bed

Locality	Strike/Dip	n	In-Situ		Tilt-Corrected		k	α_{95}
			D	I	D	I		
A Component								
Maoshan	40/50	8	324	-40	321	10	20.2	14.5
Jikang	230/51	5	12	55	348	14	71.3	9.9
B Component								
Maoshan	40/50	6	344	15	341	60	20.3	11.6
Jikang	230/51	3	336	18	344	-34	19.8	28.4

Recalculated from *Zhai* [1984], location: 31°N, 120°E.

TABLE 4. Summary of Remagnetization Data From Nanjing Area

Component	n	In-Situ		Tilt-Corrected		α_{95}	Precision	
		D	I	D	I		K2	K2/K1
A	21	[16	61]	352	12	6.5	24.6	2.51
B	17	353	12	[358	-48]	8.4	3.3	0.21

Recalculated from *Zhao* [1987], location: 32°N, 119°E. Directions thought not to represent ancient directions are given in brackets.

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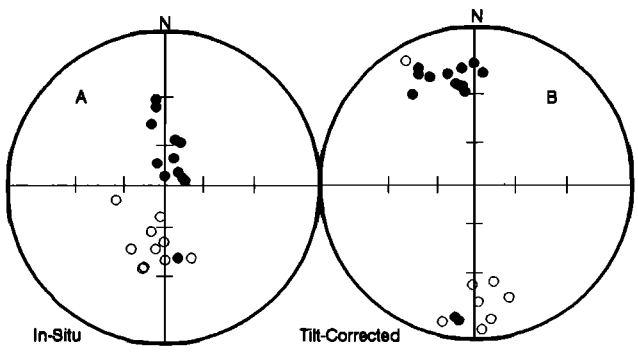


Fig. 11. Equal-angle projections of the sample directions of component A, as revealed in the Paleozoic rocks from the areas near Nanjing City, before (left) and after (right) correction for the tilt of the strata. Data are reported by *Zhao* [1987]. Conventions as in Figure 7.

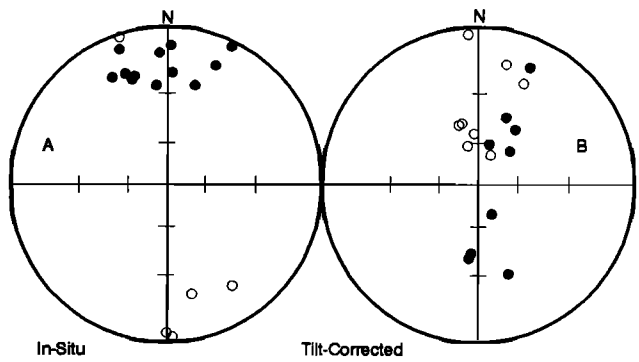


Fig. 12. Equal-area projections of the sample directions of component B, as revealed in the Paleozoic rocks from the areas near Nanjing City, before (left) and after (right) correction for the tilt of the strata illustrating that this component B fails the fold test. Data as reported by *Zhao* [1987]. Conventions as in Figure 7.

pre-folding, component B fails the fold test. Since we have observed A and B together in the same sample, at least one of them must be secondary. We argue, in fact, that both are secondary and of Triassic-Jurassic age. It should be emphasized that the area studied is adjacent to the eastern segment of the suture zone between the South and the North China Blocks, i.e., the Dabi mountains. Collision of these two blocks is thought to have happened during the Late Triassic or the Early Jurassic [*Zhang et al.*, 1984; *Yang et al.*, 1986], which conforms to the inferred folding age of the rocks studied. Fluids expelled tectonically from orogenic belts [*Oliver*, 1986], i.e., the Qinling-Dabi orogenic belts in this particular area, may have migrated through this area. Therefore, this early Mesozoic remagnetization

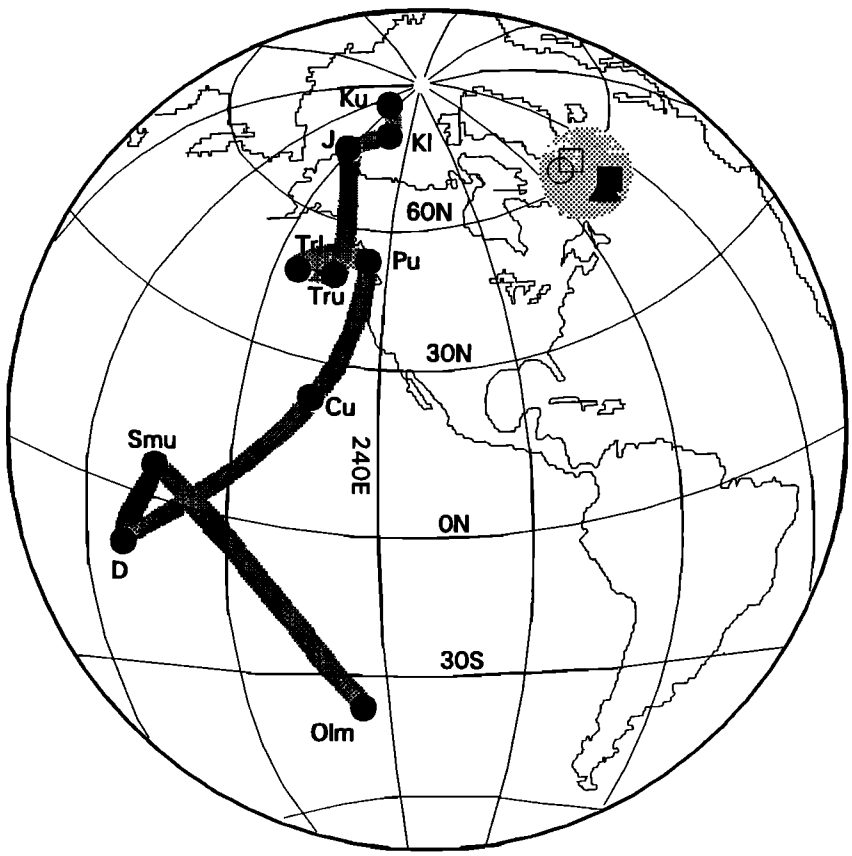


Fig. 13. Tentative Apparent Polar Wander Path for the South China Block. Paleopoles are averaged from individual results if two or more studies have been published. Ku, Late Cretaceous [*Kent et al.*, 1986]; Kl, Early Cretaceous [*Lin*, 1984]; J, Jurassic [*Lin*, 1984]; Tru, Late Triassic [*Opdyke et al.*, 1986]; Trl, Early Triassic [*Chan et al.*, 1984; *Lin et al.*, 1985; *Opdyke et al.*, 1986, *Steiner et al.*, 1989; *Heller et al.*, 1988]; Pu, Late Permian [*McElhinny et al.*, 1981; *Zhang*, 1984; *Huang et al.*, 1986; *Steiner et al.*, 1989; *Zhao et al.*, 1987]; Cu, Late Carboniferous [*Lin et al.*, 1985]; D, Devonian [*Fang et al.*, 1989*a*]; Smu, Middle/Late Silurian [*Opdyke et al.*, 1987]; Olm, Early/Middle Ordovician [*Fang et al.*, 1990]. Paleopoles of the remagnetization directions from previous and our studies are plotted in the shaded area. Solid triangle, this study, calculated from Table 2; Solid square, calculated from Kent et al's directions revealed in Carboniferous, Permian and Triassic rocks; Solid circle, calculated from Table 3 [after *Zhai*, 1984]; Open square; calculated from Table 4 [after *Zhao*, 1987].

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episode is possibly associated with hydrothermal, or perhaps hydrocarbon migration in the diagenetic history of the rock units. More compelling evidence for linking the remagnetizations with fluid activity will certainly depend on further rock magnetic as well as magnetic petrological investigations.

CONCLUSIONS and RECOMMENDATIONS FOR FUTURE WORK

1. Two remagnetization components A and B that are prefolding and postfolding respectively have been observed in the Tangshan section of South China and seem to be roughly of the same age, i.e., Triassic-Jurassic.

2. Directions similar to those of the A and B components in our study have been shown in previous studies in the east part of South China, adjacent to the Tangshan section.

3. The unblocking temperatures indicate that both magnetite and hematite are the carriers of components A and B.

4. Minor rotations are possible in the areas discussed; apparent rotations may have been introduced by steep dips or lightly plunging folds.

5. The northerly and shallow Declination/Inclination, observed in our folding-induced remagnetization directions, has also been observed in Paleozoic and Mesozoic rocks in other areas in the South China Block, such as the directions revealed from the Meishucun Precambrian/Cambrian boundary section [*Fang et al.*, 1989*b*] and directions from Triassic rocks in the Guizhou and Sichuan Provinces [*Chan et al.*, 1984; *Opdyke et al.*, 1986; *Wang and Van der Voo*, 1991]. A reevaluation is necessary for these studies to see whether some of them are remagnetizations as well.

6. In addition to the directions mentioned above, almost all directions obtained from the Paleozoic rocks in the South China Block uniquely have shallow inclinations [*Fang et al.*, 1989*a*; *Huang et al*, 1986; *Opdyke et al.*, 1987]. Because of the likelihood of relative rotations, it is thus also necessary to evaluate the possibility that remagnetizations have occurred in rocks that yielded shallow inclinations and more easterly or westerly declinations. Renewed efforts are needed to acquire more data and to conduct rock magnetic studies to verify the origins of magnetizations residing in the Paleozoic and Mesozoic rocks in South China.

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